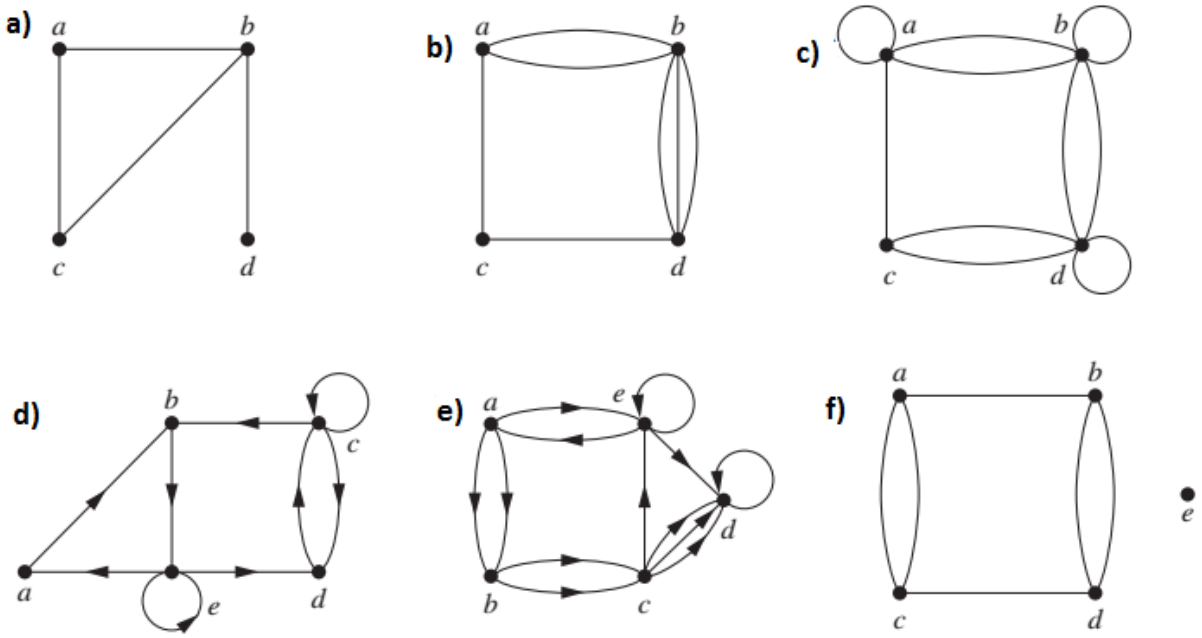
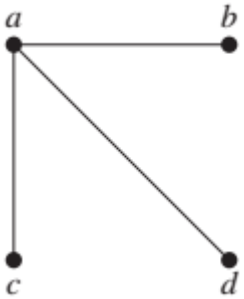
CSE2023 - ASSIGNMENT #7

Solve the following questions from course book. (Discrete Mathematics and Discrete Mathematics and Its Applications, 7th Ed. by Kenneth Rosen)

1) Determine whether the graph shown has *directed or undirected edges*, whether it has *multiple edges*, and whether it has *one or more loops*.

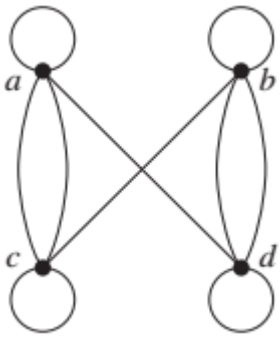


2) Draw all subgraphs of this graph.



one vertex is also a subgraph and empty set

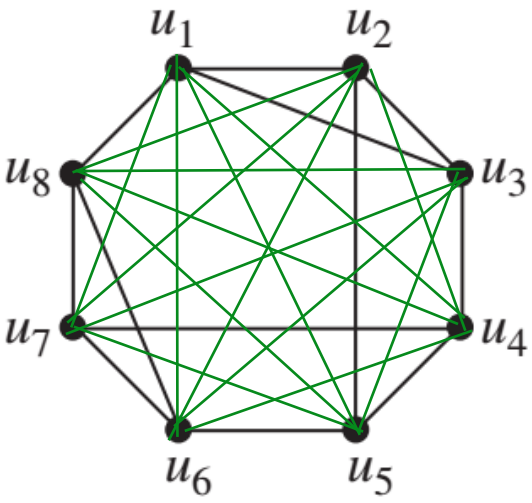
3) Represent the given graph using an adjacency matrix.

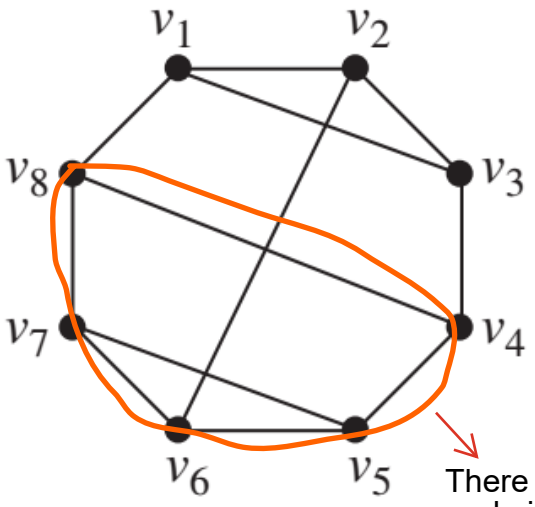


4) Draw an undirected graph represented by the given adjacency matrix.

$$\begin{bmatrix} 0 & 1 & 3 & 0 & 4 \\ 1 & 2 & 1 & 3 & 0 \\ 3 & 1 & 1 & 0 & 1 \\ 0 & 3 & 0 & 0 & 2 \\ 4 & 0 & 1 & 2 & 3 \end{bmatrix}$$

5) Determine whether the following two graphs are isomorphic or not.





There is no 5-vertices cycle in the left graph

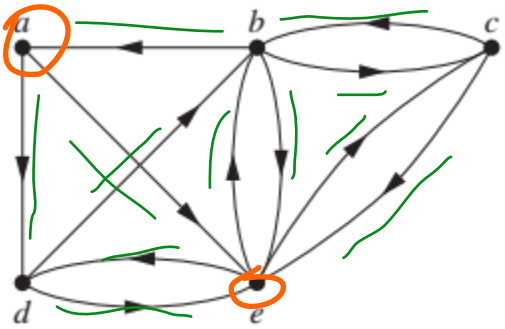
G

Edge count, same
Vertex count, same
Degree's of each vertex, same
So look at the cycles

H

Isomorphic

6) Determine whether the directed graph shown has an Euler circuit. Construct an Euler circuit if one exists. If no Euler circuit exists, determine whether the directed graph has an Euler path. Construct an Euler path if one exists.



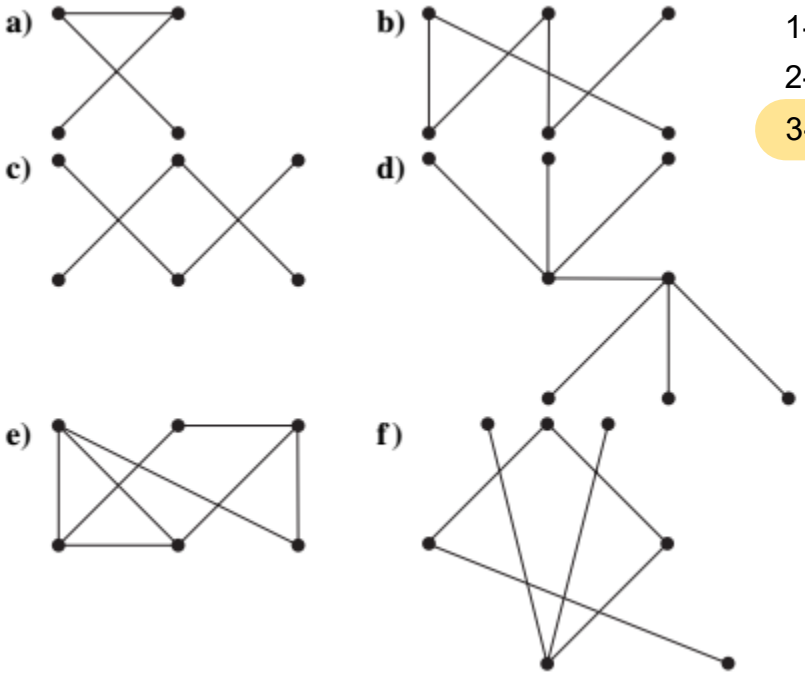
1 in 2 out, we can not get the second out-edge

Euler circuit not exists

Euler: All edges
Hamilton: All vertices

Euler Circiut: All nodes should have equal number of in and outs.
Euler Path: All nodes should have equal in and outs except only 2.

7) Which of these graphs are trees?

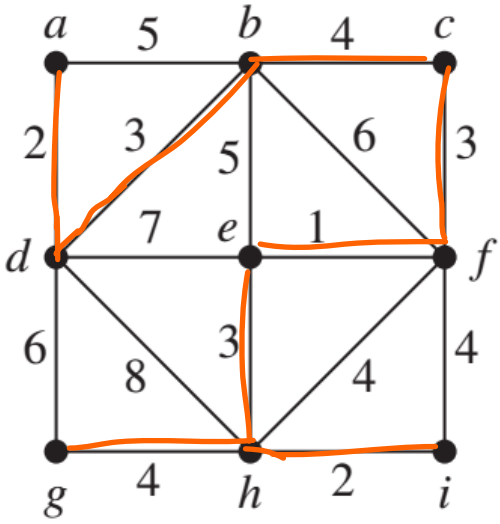


1- n vertices then, n-1 edges

2- undirected graph

3- no simple circuit

8) Use Prim's algorithm to find a minimum spanning tree for the given weighted graph.



1- Choose the smallest edge incident to a selected edge
2- Do not create cycles

Kruskal

Select the smallest edge in the WHOLE graph
Do not connect two selected vertices (Do not create cycles)